

**How do manufacturing firms evaluate the financial trade-offs of
investing in automation and digital technologies?**

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Abstract

Manufacturing firms across the globe invest in automation and digital tools and new technologies to reduce cost, increase their output, and make better quality products. Firms must carefully assess whether the long-term benefits of automation justify the initial costs. This paper discusses and explains how firms build this financial case. It uses evidence from recent industry reports and uses logic behind investments in automation technologies. Recent data shows how fast automation and use of robots is growing. The International Federation of Robotics stated that in the year 2023 approximately 4,281,585 industrial robots are operating in factories worldwide, and 541,302 new robots to be installed in the same year. Manufacturing Firms also invest in “digital” systems that support automation.

The paper shows that firms mainly use TCO (total cost of ownership) to calculate overall costs (equipment, software, integration, downtime, training, and cyber). Further NPV, IRR, and payback are used to calculate their returns. Since results are uncertain, a sensitivity analysis and often used in pilots projects to get better idea before scaling. Finance metrics with factory KPIs like downtime, scrap, and output, help track benefits after implementation to decide if the investment in the automation is good or bad.

Keywords: automation, digital technologies, Industry 4.0, financial trade-offs, total cost of ownership (TCO), net present value (NPV), payback period, manufacturing ROI, scenario analysis

International Federation of Robotics. (2024). *Record of 4 million robots working in factories worldwide: World Robotics 2024 report*. <https://ifr.org/ifr-press-releases/news/record-of-4-million-robots-working-in-factories-worldwide>

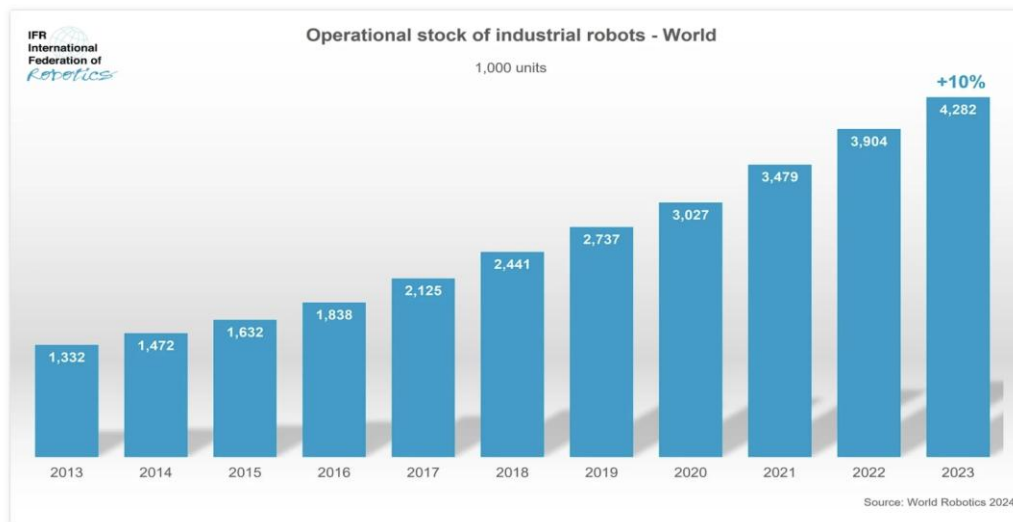
Introduction

Manufacturing firms across the globe are increasing their investment heavily in automation and digital technologies. This rapid adoption of new technology is mainly driven by strong competition, cutting down labor costs, boosting quality, and the need for faster production. Competition is driving the companies to optimise their manufacturing process. Customers now want better quality, shorter delivery times, and stable prices. Many countries are facing shortage of skilled workers and aging workforces that add to the problem. Automation and robots fill the gap and help firms handle these pressures by improving speed, accuracy, and reducing the human error.

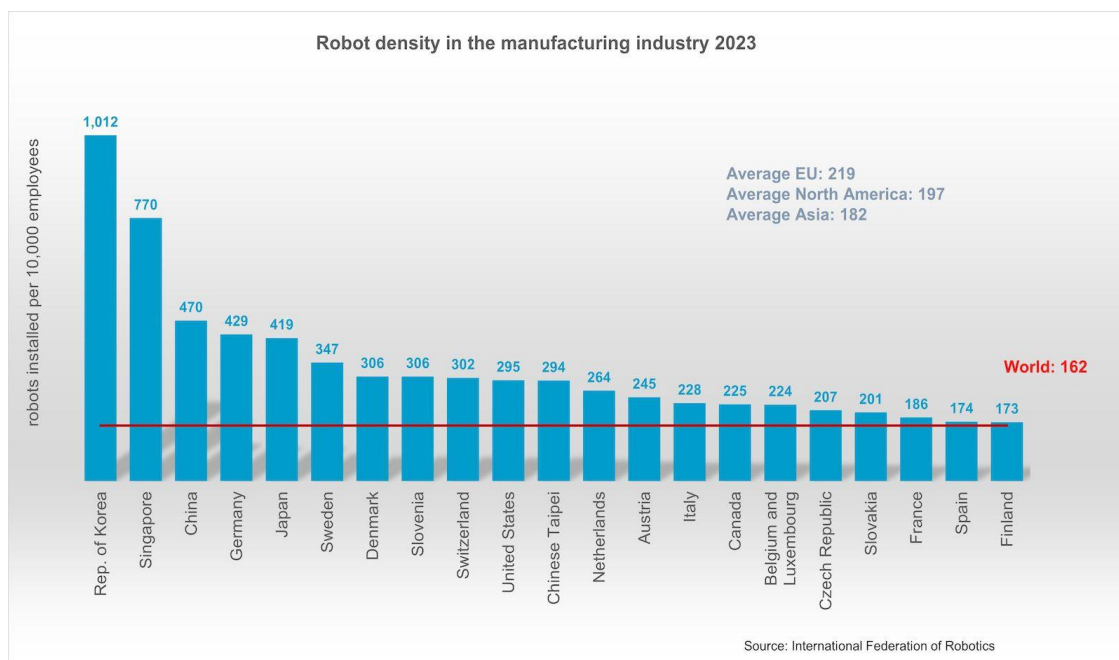
The scale of automation adoption in past decade reflects its importance. According to the International Federation of Robotics, there were more than 4.28 million industrial robots operating in factories worldwide in 2023, which is 10-12 % more than the last years. Annual robot installations have remained above half a million units in recent years. Market size for manufacturing automation has reached approximately 13 billion in 2023 and it is expected to grow to 24 billion by 2030. Automation also helps firms reduce errors, improve workplace safety, and increase output without expanding factory space. Digital technologies go beyond robots alone. Modern factories increasingly use artificial intelligence (AI), industrial internet of things (IIoT) sensors, data analytics, cloud platforms, manufacturing execution systems (MES), digital twins, and predictive maintenance tools.

These systems collect and analyze real-time data from machines and processes. A recent Deloitte smart manufacturing survey shows that firms are prioritizing investments in tools such as data analytics, cloud systems, AI, and connected sensors to support automation and decision-

making. Digital tools allow firms to create smart factories and respond quickly to changes in demand. Despite these benefits, automation and digital projects are costly and risky. Firms must spend large amounts on equipment, software, integration, training, and cybersecurity. The results are also uncertain. Some projects deliver strong savings, while others fail to meet expectations. As a result, firms must carefully compare costs with expected value.



Worldwide operational robots reach 4 million in 2023 © World Robotics 2024



The research question of this paper is:

How do manufacturing firms evaluate the financial trade-offs of investing in automation and digital technologies? This paper aims to build a structured framework that explains how firms measure costs, benefits, and risks when making these investment decisions. In particular, it examines how firms incorporate capital expenditure (CapEx), operating expenditure (OpEx), total cost of ownership (TCO), and return measures such as payback period into their evaluation processes.

In this paper, automation refers to machines or systems such as industrial robots, collaborative robots (cobots), automated guided vehicles (AGVs) for moving goods, and automated production lines. Digital technologies refer to AI, data analytics, IIoT, cloud computing, digital twins, MES, and predictive maintenance systems that support and optimize automated operations. The term financial trade-off means the balance between high upfront capital expenditure (CapEx) and long-term value, technology failure and risk. This study focuses only on financial evaluation methods, such as cost analysis, return measures, and risk assessment. It does not examine ethical issues, labor politics, or social impacts in detail, except where they affect financial outcomes or job shifts.

Literature and Concept Base

This section aims to review key ideas from various research studies of finance and operations to explain how manufacturing firms make smart choices and evaluate the financial trade-offs of investing in automation and digital technologies. Present studies show that firms rely on various methods. For instance they combine financial appraisal tools with operational thinking and risk analysis.

Manufacturing firms commonly use standard capital budgeting tools to evaluate the benefits of automation and digital projects. The most widely used methods include Net Present Value (NPV), Internal Rate of Return (IRR), payback period, and discounted payback period. NPV measures the difference between the present value of expected future cash flows and the initial investment. A positive NPV shows that the project is expected to add value to the firm so it is beneficial. IRR shows the discount rate at which NPV becomes zero and is often compared with the firm's required rate of return. It shows return in percentage but can be misleading as there can be multiple IRRs in odd cash flows. These tools are recommended in finance literature because they consider the time value of money (Brealey, Myers, & Allen, 2020).

However, many manufacturing firms use a payback **period method**. It ignores the time value. Surveys and industry reports suggest that managers prefer payback despite flaws because of its simple, easy to grasp & explain, and useful in uncertain environments (McKinsey & Company, 2023). Automation projects often involve new technologies and uncertain outcomes. Managers mostly look for quick recovery of cash and try to reduce long-term risk. Discounted payback improves this method slightly by including discounting, but it is still limited.

Each method has flaws. For instance Payback methods ignore cash flows after the cut-off period and do not measure total value created in the long run. As a result sometimes firms reject automation projects that have high long-term benefits but slower rate of returns.

NPV and IRR assume steady flows, but automation brings uneven benefits. IRR can give misleading results when projects differ in size or timing of cash flows. NPV is considered the strongest method, but it relies on assumptions about future savings, demand, and costs. As a

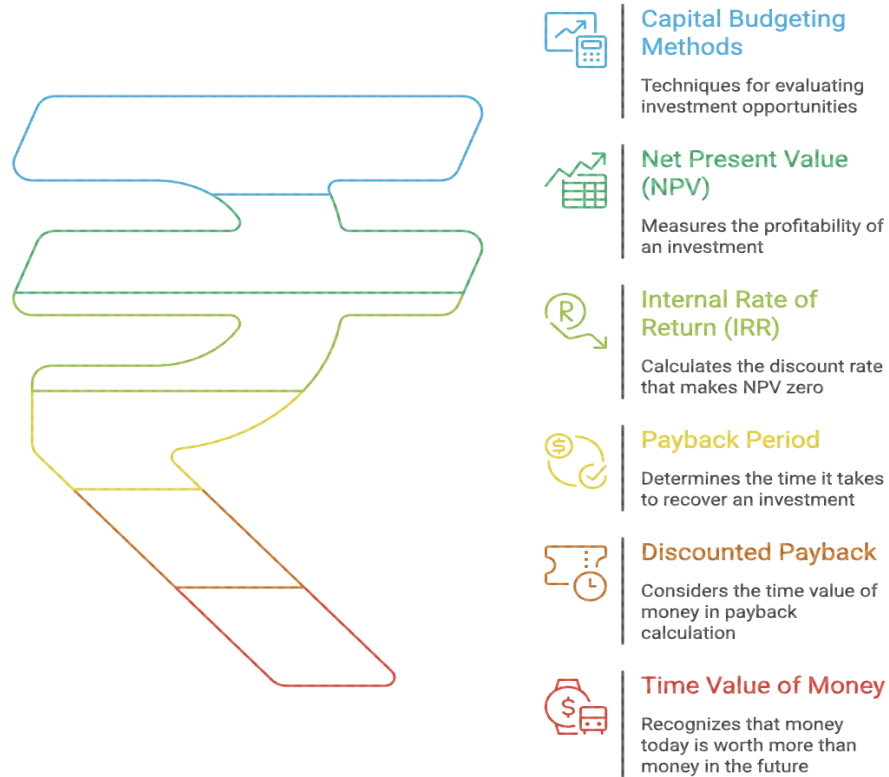
result, firms often use several methods together rather than relying on one single indicator to assess the cost benefit.

Since automation and digital systems have hidden costs such as purchase price, integration, downtime losses, training, maintenance, software licenses, cyber protection, and upgrades, firms increasingly use **Total Cost of Ownership (TCO)** and lifecycle costing(OECD, 2023). For physical automation such as robots or AGVs, integration and maintenance costs can be as large as the initial equipment cost.

TCO is even more important for digital technologies. Digital systems often rely on subscriptions, cloud services, data storage, analytics tools, and regular software updates. These are continuous costs that a system needs for its entire life time and it can increase as data volumes grow. Studies on digital transformation show that firms often underestimate expenses of data preparation, system integration, and cybersecurity (Deloitte, 2023). If these costs are ignored, the financial cost and the benefits look much stronger than it actually is. Subscriptions sometimes cost up to 60-70 % of TCO. A Deloitte study found that TCO reveals 20-30% extra costs in smart factories.

Lifecycle costing helps firms to compare alternatives over their entire life cycle. For example, a cheaper system may have lower upfront cost but higher maintenance and upgrade expenses. Studies suggest that long term cost increases for more expensive systems however they may offer better reliability and flexibility. By using the TCO method firms can better understand the true financial trade-off between short-term savings and long-term value.

Comprehensive Capital Budgeting Overview



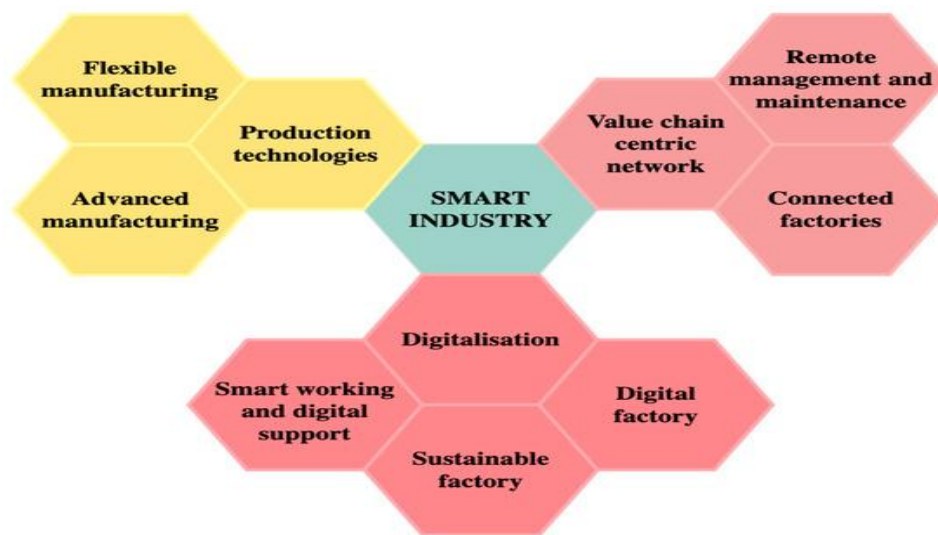
Literature also shows that the money value of automation and digital technologies that manufacturing firms invest goes beyond just direct labor cost savings. Firms measure benefits such as more productivity, improved quality, lower waste of resources, higher throughput, better overall equipment effectiveness (OEE), energy savings, improved safety, and greater flexibility. These benefits often materialise across multiple processes rather than in a single area of the factory.

Smart factory case studies suggest and highlight that digital technologies support real-time monitoring of process, predictive maintenance, and better planning that helps to reduce

unplanned downtime and improve asset utilisation (World Economic Forum, 2023). Consulting and industry reports often document improvements of 5-20 % in OEE along with significant reductions in defects and energy use when digital systems are used (McKinsey & Company, 2023). Although these gains are harder to convert and calculate into cash flows, they strongly influence competitiveness and resilience of the firm.

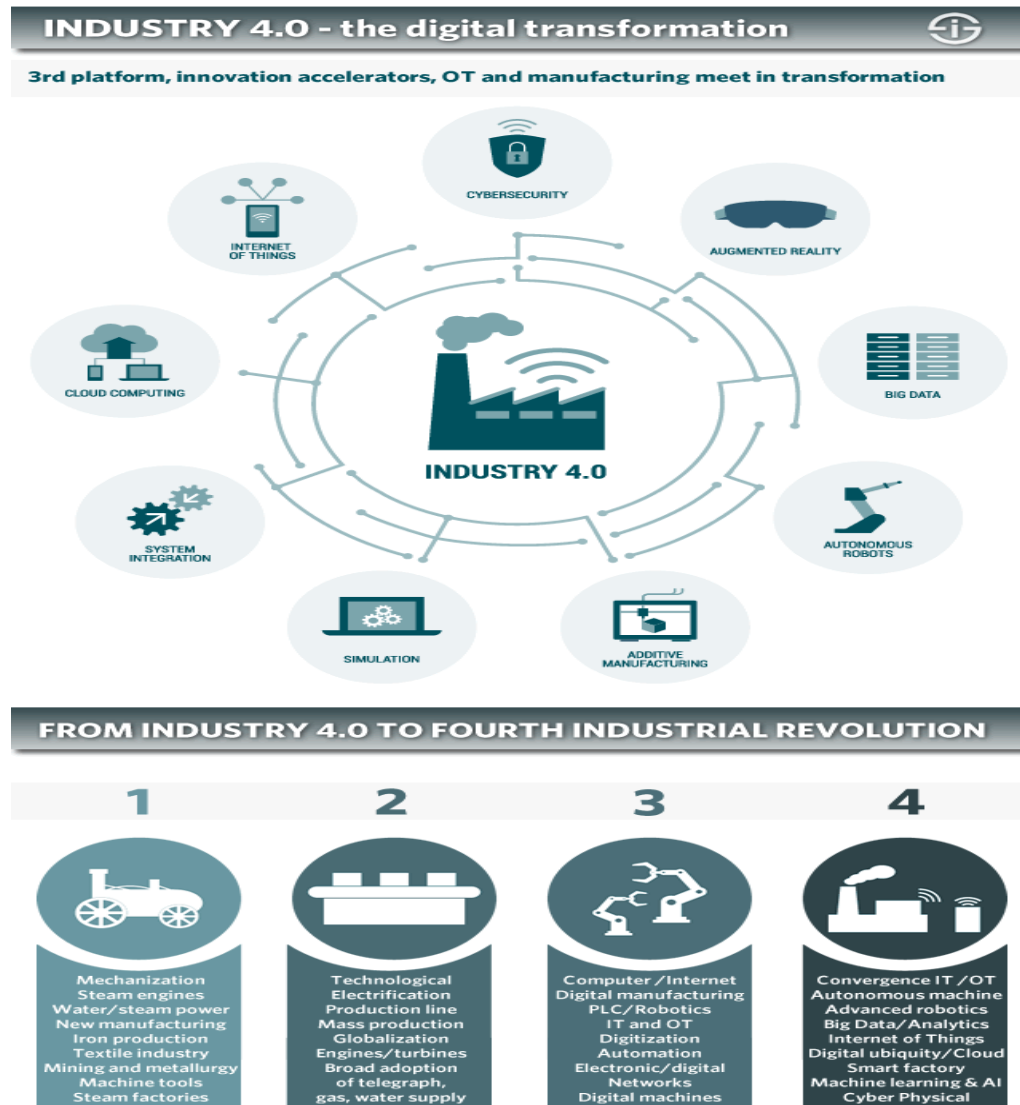
Automation and digital investments come with high uncertainty and risks such as technology risk, integration risk, adoption risk, skills shortages, and cybersecurity threats. Systems may not perform as expected, workers may resist change, or firms may lack the skills to use data effectively. Cyber risks are especially important for connected factories.

As a result of these risks, literature emphasizes risk-adjusted evaluation. Firms use scenario analysis and sensitivity analysis to test how changes in assumptions affect results. This helps decision-makers understand downside risks and avoid over-optimistic forecasts. Overall, the literature agrees that strong financial evaluation combines cash-flow analysis, TCO, operational metrics, and explicit risk assessment.



Source - Hetmańczyk, M. P. (2024). *A method for evaluating the maturity level of production process automation in the context of digital transformation—Polish case study*. **Applied**

Sciences, 14(11), 4380. <https://doi.org/10.3390/app14114380>



Source- De Clerck, J.-P. (2016, February 5). *Industry 4.0 and the fourth industrial revolution explained*. i-SCOOP. <https://www.i-scoop.eu/industry-4-0/>

Analytical Framework

This section presents a structured framework discussing how manufacturing firms evaluate the financial trade-offs of investing in automation and digital technologies. The framework discussed here draws on common practices identified in industry reports, finance literature, and smart manufacturing studies.

It highlights that firms combine their financial analysis and investment required with operational thinking and risk assessment to make a final decision. Some of the most common problems that manufacturing firms are facing now are on high machine down time, defect rates in the products, shortage of labour, safety risk of workers and very long lead time. Before investing in new technology the firms identify these key problems.

Once the main issues are identified the firms look for the possible solutions and compare the options available with them. Possible solutions could be physical automation such as utilising industrial robots or automated guided vehicles (AGV) or Digital solutions such as upgrading the manufacturing execution systems (MES) using analytic software or predictive maintenance tools.

Comparing different options also allows the firms to see which tool is best for them in terms of cost and in terms of solving their problem. Firms measure current performance using data such as operating cost, labor hours, scrap rates, lead time, power consumption, and overall equipment effectiveness (OEE) to measure these. Firms identify the baseline to compare the relative performance and the future benefits of the technology. Selection of a good baseline is important because a poor baseline data can lead to inaccurate and unreliable financial estimates.

Once the baseline is calculated, firms **estimate expected future benefits**. These add gains such as labor savings, reduced scrap, higher throughput, and lower downtime. Firms also consider soft or indirect benefits such as improved quality consistency, safer working conditions, and faster response to demand changes.

After the full estimation of cost and the benefits the firms run the financial appraisals. Common tools include net present value (NPV), internal rate of return (IRR), and payback period. NPV is often preferred because it shows whether the project adds value after accounting for the time value of money. Firms use mostly payback method to assess the liquidity risk and to assess the speed of recovery. As the outcomes are not certain firms are at risk as well hence they indulge in a best case and worst case scenario analysis.

Sensitivity analysis helps the firms to identify and understand the downside risk and avoid the too much dependence on optimistic decisions as it helps to identify the impact of assumptions in the result.

Finally manufacturing firms make a clear investment plan to take decisions and create a KPI tracking plan. This plan reflects the clear performance indicators before the final implementation of the technology.

After this the technology and the digital tools are deployed and KPI are clearly tracked to check if the promised benefits are actually being delivered or not. This helps to improve the future investment decisions. The entire evaluation Framework includes all types of cost including the hidden cost along with the visible cost. The visible cost includes Robots, cobots, sensors, PLC upgrades software systems like MES or expenses for the integration work external counselling etc.

Studies have found that many automation project fail to meet the desired outcomes and expected returns as the firms underestimate the hidden cost or sometimes ignore these hidden costs. These costs include the training expenses for workers ,the losses that firms incur during the installation and testing, the time that is taken during the data cleaning and preparation of digital systems.

Adoption of Digital technologies also require regular updates and system support. Deloitte (2023) highlights that these Direct costs or significant does it should be included in the overall cost while calculating the return and the benefits .

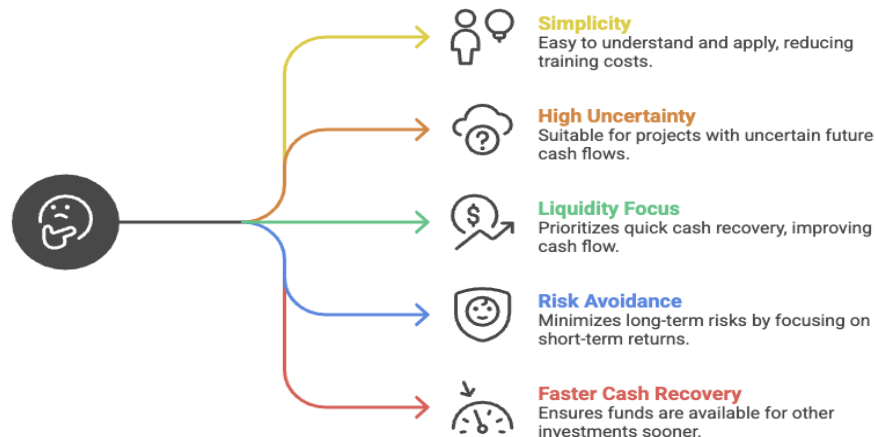
Benefits from automation and digital investments can be divided into **direct** and **strategic** benefits. Direct benefits are easier and simple to measure as they are clearly visible . They include labor cost reduction, scrap reduction, waste reduction, higher throughput, reduced unplanned downtime, and lower energy consumption. These benefits directly affect operating costs and margins.

Strategic benefits are harder to calculate and quantify but still important as they provide many benefits to the firms for instance faster changeovers, better demand forecasting, higher process flexibility, improved regulatory compliance, stronger customer service, and greater supply-chain resilience.

To remain competitive in the long run rather than looking at the benefits of short run the strategic benefits must be included and calculators. Recent studies suggest that firms increasingly value digital benefits. A Deloitte smart manufacturing survey reports the automation investment technology and the digital tools help firms to improve their decision quality and operational visibility as well. Hence many firms prefer to invest in Data Analytics, cloud platforms,

Artificial Intelligence and **IIoT** (Deloitte, 2023). Including both direct and strategic benefits helps firms make balanced decisions that reflect true long-term value.

Should the firm use the payback period method for capital budgeting?



Financial Trade-off Analysis

Most manufacturing firms typically use discounted cash flow analysis as it helps the organisation to compare the value of expected future benefits with the capital invested today. The primary outcome of discounted cash flow method (NPV) is net present value. When the NPV calculated is positive, it means that the project will add value to the firm in the long run. Consider a medium-sized manufacturing firm planning to invest in an automated production cell supported by digital monitoring software. The initial investment is ₹10 crore which includes robots, sensors, integration, and software licenses. The life of the system is expected to be 8 years. In this particular case the firms calculate the expected benefits in the form of cost that their

saving by reducing the labours reducing the wastage and unplanned down time. The firm estimates annual gross savings of approximately ₹2.5 crore, arising from reduced labour costs, lower material wastage, and decreased unplanned downtime. At the same time their operating and maintenance cost is about ₹0.5 crores per year for power consumption, maintenance subscription and IT support. This results in a net annual cash inflow of ₹2 crore. To evaluate the project, the firm discounts these cash flows using its weighted average cost of capital (WACC) or hurdle rate. Manufacturing firms usually use rates between 8%-12%, depending on risk. If the firm uses a discount rate of 10%, the present value of future cash inflows can be compared with the initial investment. If the NPV is positive, the project is financially acceptable.

Firms also calculate the total payback period, which helps them to see how long it takes to recover the initial investment from net cash inflows. In the example above, the payback period is slightly over five years. This shows the trade-off between long-term value and short-term liquidity. Finance literature and industry practice agree that NPV gives a more complete picture than payback, but firms still use both together (Brealey et al., 2020; McKinsey & Company, 2023).

Automation outcomes differ and give different outcomes across factories. For this reason, firms rarely rely on one set of assumptions. Instead, they use sensitivity analysis (best case and worst case) to test how changes in key variables affect NPV. This helps managers understand which assumptions matter most. Common variables tested include downtime reduction percentage, scrap reduction rate, wage inflation, technology adoption rate, and integration time. For example, if downtime reduction falls from an expected 20% to only 10%, annual savings may drop very quickly potentially reducing NPV or even making it negative. Similarly, if wages

of workers rise faster than expected, labor savings become more valuable, increasing NPV to become positive.

Sensitivity analysis mainly shows that reduction in downtime and adoption rate have the largest impact on value created. If workers do not fully use the system, expected benefits fail to materialize and give better return to the firms. Studies of smart manufacturing projects show that performance gains depend heavily on how well systems are integrated and used, not just on the technology itself (Deloitte, 2023). This analysis is critical because automation projects are uncertain by nature. Sensitivity analysis helps the firms to identify “break-even” points and focus management attention on the most important risks. It also encourages more realistic and practical forecasts and reduces over-confidence in best-case outcomes and scenarios.

It has been found that most firms use at least three different scenarios such as conservative, base and optimistic because scenario analysis looks at the combinations of changes across multiple variables while sensitivity analysis looks at the changes caused by one variable at a time. Conservative scenarios usually assume higher costs than expected and delayed benefits. For instance due to higher maintenance cost and slow learning the savings might get delayed. In the base scenario as the expectations are based on the past projects the assumptions are more realistic and practical. Optimism scenario assumes higher benefits as we assume the system to be stable and rise in demand. Furthermore, many firms also consider failure scenarios. In this scenario the system integration is usually delayed as the workers resist change because of the field that they might lose their job or the data quality is very poor. As a result the gains and the benefits always remain less and the cost goes high. World economic forum and several industry reports highlight that digital transformation projects failed to meet there targets and expected benefits because of the weak management control and skills gaps (World Economic Forum,

2023). Scenario analysis also helps the management and decision makers to understand the risk that is involved in the down side of the firm. Even if the base case in NPV is positive, a large negative outcome in the failure scenario may cause the firm to delay and re-design the entire project. This approach mainly supports balanced decision making specially in case of large and irreversible investments.

To manage uncertainty, firms also use risk-adjusted evaluation tools. One approach is to increase the discount rate for risky projects of the firms. A higher discount rate reduces the present value of future cash flows and reflects higher uncertainty. Another approach is to use probability-weighted cash flows, where each scenario is given and assigned returns and expected values which are calculated based on the assigned likelihood .In recent few years firms have shifted their approach to first try technology on pilot pieces and then scale it later if it works for them. Instead of committing full capital upfront, firms initially invest in a small pilot project and scale it only after seeing if the targets are met and expected returns are achieved in the pilot project. This reduces downside risk and allows learning before large investments are made. The OECD notes that this approach is especially useful for digital technologies, where learning effects are strong and uncertainty is high (OECD, 2023).

Due to rising uncertainty, many manufacturers now prefer shorter payback horizons, often between one and three years for automation projects. This does not mean that firms ignore long-term value. Instead, they seek early cash recovery while still tracking long-term strategic benefits. This reflects a careful balance between growth, risk control, and financial discipline.



Organizational and Strategic Factors Affecting the Business Case

The success of automation and digital investment depends on factors including financial models. This is evident in industry wide research and practices in their policies regarding organisational readiness, governance, strategic suitability. Projects with positive NPV may also fail if firms do not prepare themselves, implement and manage up to date systems. There is a skill gap in human resources. Automation and digital systems need programmers, data analyzers, maintenance as well as connected equipment. Neglecting these prerequisites may result in underutilization of capacity and lower than expected benefits. Several industry reports suggest that the primary barrier to successful transformation to automation is lack of required skill set (Deloitte, 2023). Data maturity represents another critical constraint. Firms with fragmented IT systems, inconsistent data standards and poor data quality struggle to produce reliable insights. As a result, advanced digital systems may fail to improve decision-making or operational

performance. Advanced machines require proactive and condition based maintenance, failing which may lead to increased downtime and operational underperformance.

Strategic risks also arise from over dependence on vendors for complex and highly customized systems. These systems lock firms into a long-term relationship with suppliers for upgrades and problem solving raises long term risk and costs. Studies are proof that process and people along with technology deliver expected deliverance. Proper training, standard operating protocols and inter and intra function collaboration investments generate better financial outcomes than dependence only on equipment (OECD, 2023). Strong governance significantly aids in improved financial returns. Ambiguity in responsibility is a major factor why benefits are not fully materialized. As far as the question of ownership of return on investment is concerned, successful firms share ownership across operations, finance and IT.

To track performance after implementation of policies, employing KPI dashboards apart from other tools is very common. These dashboards have several indicators including scrap rate, downtime, throughput, energy use, labour productivity. Regular monthly or quarterly reviews help firms assess whether expected benefits are being realised. A benefits realization plan links each KPI to assign accountability and financial gains. Research on previous such practices show that firms with active performance tracking practices are better placed to achieve projected returns than those who do it partially. (McKinsey & Company, 2023). Considering strategic value to the company over the long term compels some firms to invest in automation and digital technologies even when short term NPV is modest. Benefits such as customization, increased flexibility, client support and improved resilience from supply chain disruptions may not appear fully in initial cash flows but in the long term may improve competitiveness. Examples from advanced manufacturing sites often called “light house” factories, show this effect without any

doubt. The World Economic Forum reports that such factories quite often achieve sustained productivity gains and reduced cost over time, even when initial investment were large (WEF, 2023).

Discussion

This analysis shows that the best manufacturing firms often synchronise various risks, strategies, and decision making for investment in automation and digitization rather than basing their decisions on a single financial figure. The integrated approach enables them to make calculated and less risky steps thereby avoiding costly mistakes. Relying on several indicators such as NPV (Net Present Value), IRR (Internal Rate of Return), TCO (Time Cost of Ownership) offers multiple competitive strengths and integrates sustenance of firms. (OECD Organization for Economic Cooperation and Development, 2023)

Financial indicators along with operational performance paints a better picture. Improved operational data is usually followed by better financial results. Strong risk perspective accompanies other financial metrics in any good evaluation. Every new adoption involves a certain degree of risks therefore strong risk assessment and scenario analysis, sensitivity analysis under varied practical like conditions is evaluated by firms. Deloitte (2023) highlights that many projects fail not because the technology is inadequate, but because organisational processes and workforce capabilities are insufficiently prepared.

Beyond short-term financial returns, successful firms also evaluate the strategic value of automation investments. Considerations such as long-term flexibility, preparedness for supply chain disruptions, system scalability, and workforce upskilling play a central role in investment

decisions. Further, periodic review of performance improvement, compatibility of system and cyber security risks is essential for promised returns.

Conclusion and Recommendations

This paper presents how manufacturing firms evaluate tradeoffs of investing in automation and digital technologies by employing a combined and structured approach. High upfront capital costs, long term benefits are tested against financial tools such as NPV(Net Present Value), IRR(Internal Rate of Return), TCO(Time Cost of Ownership). Financial results and operational performance are linked for a clearer image. Scenario analysis, sensitivity analysis and phased investment approach are some of the methods for risk analysis and generate expected returns. For resilient and contingent requirements firms use multiple metrics rather than a single indicator. Structured and disciplined approach can improve the quality of their investment. A pilot and stage wise approach can add another layer of protection against risks. Small pilot projects for testing and scaling only can be initiated after clear demonstration of benefits.

Future research can examine how financial evaluation differs across manufacturing sectors such as automotive, electronics, and pharmaceuticals. More work is also needed on how small and medium-sized enterprises evaluate automation under capital constraints, and how national policies and incentives influence investment decisions.

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